

SLIDING WINDOW.

1) $K = 2$ | $K = \infty$.

2) Consecutive | sub array | sub string.

Buzzword

$K = 2$

2	8	7	12	14
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$i = 0$

$j = 0$. two starting variables.

$j = 1$ $j = 2$ $j = 3$.

V.V. Imp

① $i = 0$

$j = 0$

② If window size not achieved $\rightarrow$ move j ($j++$)

③ If we achieve the window size
 $\rightarrow$ maintain the window size

$\rightarrow i++$
 $\rightarrow j++$

④ Remove the visited Element

⑤ Add the first Element

Template for Sliding Window:

$i = 0$

$j = 0$

min-size = 0

max

check for window size.

if $(j - i \neq k)$:

→ Add j^{th} Iteration

$j = j + 1$

if size achieved $(j - i + 1 == k)$

Math Operation

$i + 1$

→ Remove

$j + 1$

i^{th} Iteration

class Solution:

def maxSubarraySum(self, arr, k)

$n = \text{len}(arr)$

$i = 0$

$j = 0$

min-sum = 0

max-sum = 0

while $j < n$:

 window-sum += arr[j]

 if $j - i + 1 < k$:

$j + 1$

 elif $j - i == k$:

~~max = max~~

 max-sum = max(max-sum, window-sum)

 window-sum -= arr[i]

$i + 1$

$j + 1$